Supplementary Material

QSMEF: A Methodological Framework for Functional Analysis of Quantum Implementations

Applicability Analysis of Quantum Phase Estimation (QPE) in Shor's Algorithm

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Abstract

This supplementary document complements the paper *QSMEF: A Methodological Framework for Functional Analysis of Quantum Implementations* with an applicability analysis of the Quantum Phase Estimation (QPE) subroutine used in Shor's algorithm.

The initial experiment applied the QSMEF procedure to a decomposition of QPE into state preparation, superposition generation, and controlled modular evolution. It also defined a Hermitian observable and constructed a cooperative game to compute Shapley values.

A subsequent methodological analysis showed that this decomposition does not preserve functional comparability among coalitions. Neutralizing some of these operations changes the computational context of the remaining ones and, consequently, the meaning of the property evaluated across configurations.

This analysis identifies a boundary of QSMEF applicability. A meaningful functional attribution requires not only a structurally compatible decomposition and a Hermitian observable, but also functionally comparable coalitions and an observable whose meaning remains invariant across them.

1 Introduction

The companion paper introduces QSMEF and evaluates it through the Shenvi–Kempe–Whaley (SKW) coined quantum walk search algorithm on the hypercube. The selected decomposition of SKW into oracle, Grover coin, and flip-flop shift supports the coalition analysis because neutralizing these components preserves the structural and functional context required for comparison.

The supplementary material originally explored Quantum Phase Estimation (QPE) in Shor's algorithm as an additional evaluation. QPE provides a different computational structure and therefore offers an opportunity to examine the scope of the framework beyond the SKW case.

The initial experiment divided QPE into state preparation, Hadamard operations, and controlled modular evolution. It then constructed coalitions by replacing absent components with the identity operator and computed Shapley values from an observable related to period estimation.

Further analysis of this construction revealed a methodological issue. The QPE stages depend on one another to generate and interpret phase information. Removing one stage can therefore alter the functional role of the operations that remain. Under these conditions, coalition values no longer describe comparable partial realizations of the same functional process.

For this reason, this document revisits the QPE experiment as an applicability analysis. The experiment helps identify the conditions that a decomposition and an observable must satisfy before QSMEF can interpret Shapley values as functional contributions.

2 Quantum Phase Estimation in Shor's Algorithm

2.1 Modeling and Architecture

QPE uses a counting register and a work register. Their joint Hilbert space is

$$\mathcal{H} = \mathcal{H}_{\text{count}} \otimes \mathcal{H}_{\text{work}}. \tag{1}$$

The counting register contains m qubits and therefore spans $Q = 2^m$ computational basis states. The modular unitary operator acts on the work register according to

$$U|y\rangle = |ay \bmod N\rangle, \tag{2}$$

where a is coprime to N. Shor's algorithm seeks the period r that satisfies

$$a^r \equiv 1 \pmod{N}. \tag{3}$$

The eigenvectors of U satisfy

$$U|u_s\rangle = e^{2\pi i s/r}|u_s\rangle, \qquad s \in \{0, \ldots, r-1\}. \tag{4}$$

Each eigenvector $|u_s\rangle$ carries a phase s/r from which QPE can recover information about the period.

The initial work-register state can be written in the eigenbasis of U as

$$|1\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle. \tag{5}$$

The QPE procedure prepares the counting register in superposition, applies controlled powers of U, and uses the inverse Quantum Fourier Transform ($\text{QFT}^\dagger$) to convert the encoded phase information into a computational-basis representation suitable for measurement.

3 Initial Exploratory Decomposition

The initial experiment considered

$$B_{\text{QPE}} = \{B_0, B_1, B_{2,0}, B_{2,1}, B_{2,2}, B_{2,3}, B_{2,4}\}. \tag{6}$$

The components represent:

- B_0: preparation of $|0\rangle^{\otimes m} \otimes |1\rangle$;
- B_1: Hadamard gates on the counting register;
- $B_{2,j}$: controlled application of U^{2^j}, for $j = 0, \ldots, 4$.

The initial construction kept $\mathrm{QFT}^\dagger$ and measurement outside the set of attributable components and treated them as part of the readout context.

This decomposition identifies recognizable circuit stages, but that property alone does not make it suitable for QSMEF. The framework constructs coalitions by replacing absent components with the identity operator while preserving the order of the components that remain. Therefore, the resulting configurations must preserve enough functional context to support a meaningful comparison.

4 Applicability Conditions

The QPE analysis motivates three conditions that QSMEF must verify before interpreting coalition values as functional contributions.

4.1 Structural Compatibility

Every coalition must operate within the same Hilbert space and support the same functional metric.

QSMEF defines this metric from a Hermitian observable H:

$$M_H(\rho) = \mathrm{Tr}(H\rho), \tag{7}$$

with

$$H = H^\dagger. \tag{8}$$

Hermiticity guarantees a real expectation value, while structural compatibility guarantees that QSMEF can evaluate the same operator for every configuration.

4.2 Functional Comparability of Coalitions

Replacing an absent component with the identity must preserve the functional context of the operations that remain.

A coalition does not need to reproduce the complete algorithm. It must, however, represent a meaningful partial configuration of the same process. If neutralization changes the role of the remaining components, the coalition no longer supports the intended comparison.

4.3 Semantic Invariance of the Observable

The observable must represent the same property for every coalition.

Dimensional compatibility and Hermiticity do not guarantee this condition. QSMEF can compare expectation values only when $M_H(\rho_C)$ retains the same functional interpretation for every $C \subseteq B$.

Together, these conditions connect the mathematical construction of the cooperative game with the functional meaning that QSMEF assigns to it.

5 Applicability Analysis of the QPE Decomposition

The initial QPE decomposition does not preserve functional comparability for all coalitions.

The Hadamard block prepares the counting-register superposition on which the controlled powers of U act. Those controlled operations encode the phases associated with the eigenvalues of U, and $\mathrm{QFT}^\dagger$ subsequently converts that information into amplitudes that support period estimation.

This dependency creates a problem for component neutralization.

Consider a coalition that excludes B_1 but retains one or more $B_{2,j}$ components. Replacing the Hadamard block with the identity changes the input state of the controlled modular operations. These operations still define valid unitaries, but they no longer perform the same functional role that they perform after the counting register enters the required superposition.

The same issue arises when a coalition removes selected controlled powers of U. $\mathrm{QFT}^\dagger$ then receives a state that encodes phase information under different conditions. As a result, the readout no longer has the same relationship with the period-estimation process across all coalitions.

The configurations remain mathematically valid quantum states, but they do not necessarily represent comparable partial executions of QPE. Therefore, their metric values cannot support a common functional attribution.

6 Observable Analysis

The initial experiment used the projective observable H_{per} on the counting register.

Let $\mathcal{K}_r$ contain the measurement outcomes compatible with the period r within a tolerance ε. The observable takes the form

$$H_{\mathrm{per}} = \left(\sum_{k \in \mathcal{K}_r} |k\rangle\langle k| \right) \otimes I_{\mathrm{work}}, \tag{9}$$

where I_{work} acts on the work register.

H_{per} is Hermitian and acts on the Hilbert space of the implementation. For the complete QPE procedure, its expectation value quantifies the probability associated with outcomes compatible with the target period.

The difficulty appears when QSMEF evaluates the same observable after neutralizing components. If a coalition changes the process that generates the phase information, the expectation value still exists, but it no longer necessarily measures the same functional property under the same computational conditions.

Thus, H_{per} satisfies the mathematical requirements of an observable but does not guarantee semantic invariance for the initial coalition construction.

7 Exploratory Cooperative Game

The initial experiment defined the characteristic function as

$$v(C) = M_H(\rho_C) - M_H(\rho_\emptyset), \tag{10}$$

where ρ_C denotes the state associated with coalition C and $\rho_\emptyset$ denotes the empty-coalition state.

For each component b, the experiment computed

$$\phi_b = \sum_{C \subseteq B \setminus \{b\}} \frac{|C|!(|B| - |C| - 1)!}{|B|!} \left[v(C \cup \{b\}) - v(C) \right]. \tag{11}$$

The resulting allocation satisfies

$$\sum_{b\in B}\phi_b = v(B). \tag{12}$$

This equality verifies the Shapley efficiency property for the constructed game. It does not establish functional comparability among the coalitions. The latter depends on the QSMEF applicability conditions described above.

8 Exploratory Numerical Results

We retain the original numerical results to document the experiment and support reproducibility.

For $N = 21$, $a = 2$, $m = 5$, and $\varepsilon = 1/2^5$, the initial construction produced the Shapley values shown in Figure 1 and Table 1.

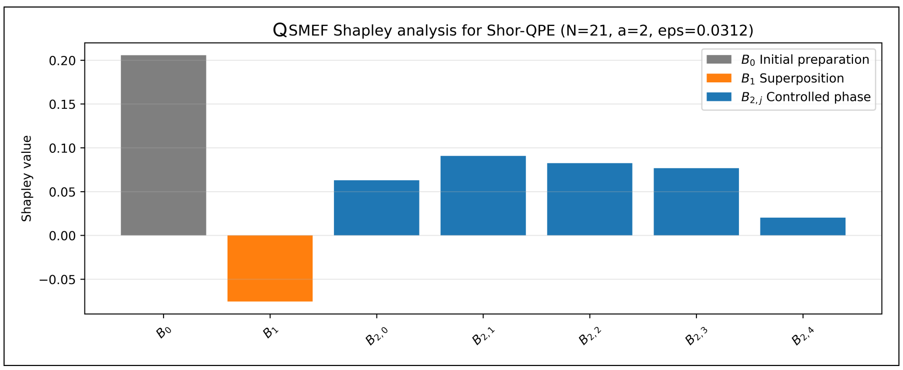

Figure 1: Shapley values obtained from the initial QPE cooperative-game construction for $N = 21$ and $a = 2$.

Table 1: Shapley values obtained from the initial QPE construction.

Component	ϕ_b	$\phi_b/v(B)$ (%)
B_0	0.205753	44.39
B_1	-0.075497	-16.29
$B_{2,0}$	0.062984	13.59
$B_{2,1}$	0.090666	19.56
$B_{2,2}$	0.082468	17.79
$B_{2,3}$	0.076805	16.57
$B_{2,4}$	0.020308	4.38
Total	**0.463487**	**100.00**

The metric values for the complete and empty configurations are

$$M_{H_{\rm per}}(\rho_B) \approx 0.745 \tag{13}$$

and

$$M_{H_{\rm per}}(\rho_\emptyset) \approx 0.281. \tag{14}$$

Therefore,

$$v(B) \approx 0.463487, \tag{15}$$

which equals the sum of the Shapley values.

These results describe the cooperative game generated by the initial construction. The applicability analysis prevents us from assigning them the stronger interpretation of functional contributions within QSMEF.

9 Exploratory Modified Implementation

The original experiment also modified component $B_{2,2}$ by replacing the unitary constructed with $a = 2$ with the corresponding operator for $a = 8$.

Figure 2 compares the Shapley values of the original and modified constructions.

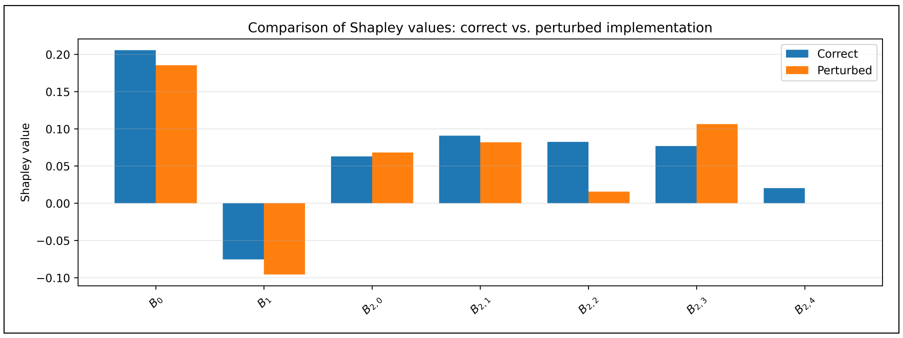

Figure 2: Shapley values obtained from the original and modified QPE cooperative-game constructions.

Table 2: Comparison of Shapley values after modifying $B_{2,2}$.

Component	**Original**	**Modified**	**Difference**
B_0	0.205753	0.185445	-0.020308
B_1	-0.075497	-0.095805	-0.020308
$B_{2,0}$	0.062984	0.068229	0.005246
$B_{2,1}$	0.090666	0.081870	-0.008796
$B_{2,2}$	0.082468	0.015579	-0.066889
$B_{2,3}$	0.076805	0.106278	0.029473
$B_{2,4}$	0.020308	0.000000	-0.020308
Total	**0.463487**	**0.361596**	**-0.101891**

The modification reduces the value associated with $B_{2,2}$ from approximately 0.082 to 0.016 and changes the allocation among the other components. The total game value decreases from approximately 0.463487 to 0.361596.

These changes show that the cooperative game responds to modifications of the underlying circuit. However, the lack of functional comparability prevents QSMEF from interpreting the differences as changes in the functional contributions of the QPE components.

10 Methodological Implications

The QPE experiment highlights the distinction between structural and functional decomposition.

A circuit can contain clearly identifiable stages without allowing their independent neutralization for functional attribution. QPE illustrates this situation because preparation, phase encoding, and phase readout derive their meaning from their interaction.

The experiment also shows that Hermiticity alone does not make an observable suitable for QSMEF. The observable must retain the same functional meaning for every configuration that enters the cooperative game.

These observations refine the applicability criteria of the framework. Before constructing the characteristic function, the analysis should verify that

1. all configurations operate within a structurally compatible Hilbert space;

2. component neutralization preserves functional comparability among coalitions; and

3. the observable retains its semantic interpretation across all coalitions.

These conditions determine whether QSMEF can interpret the resulting Shapley values as functional contributions rather than only as allocations of an abstract cooperative game.

11 Conclusions

The analysis of QPE identifies a limit to the applicability of the initial QSMEF construction.

The original decomposition separates recognizable stages of the QPE circuit, but neutralizing those stages does not preserve their functional context across all coalitions. In addition, the period observable remains mathematically well defined while its functional interpretation changes when the coalition alters the process that generates phase information.

The numerical experiment still defines a valid cooperative game and produces Shapley values that satisfy the efficiency property. However, the lack of functional comparability prevents QSMEF from interpreting those values as functional contributions of the QPE components.

This result refines the methodological scope of QSMEF. The framework requires a functional decomposition that supports meaningful neutralization and an observable that represents the same property throughout the coalition space.

The SKW quantum search analyzed in the companion paper remains the application case in which the selected decomposition and observable satisfy these requirements. The QPE experiment complements that case by showing where the methodology reaches an applicability boundary.